Experiment – 05

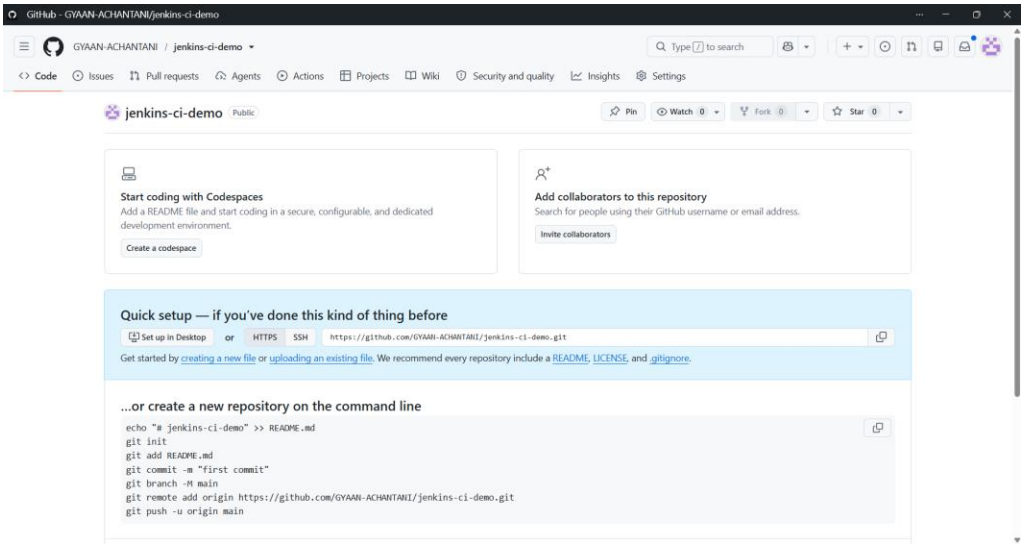
Aim:- To Automate application deployment using GitHub/Jenkins.

Theory:- Jenkins is an open-source automation server that is mainly used for Continuous Integration (CI) and Continuous Delivery/Continuous Deployment (CD). It helps developers automate the process of building, testing, and deploying software whenever changes are made to the source code. Jenkins can monitor a project repository, detect new code changes, automatically download the latest code, build the application, run test cases, and report whether the build was successful or failed. Jenkins supports many programming languages and tools and can be extended using plugins. It provides a web-based interface where users can create and manage jobs, configure build steps, view console output, and monitor the status of projects. The main advantage of Jenkins is that it reduces manual work and helps detect errors early in the software development process.

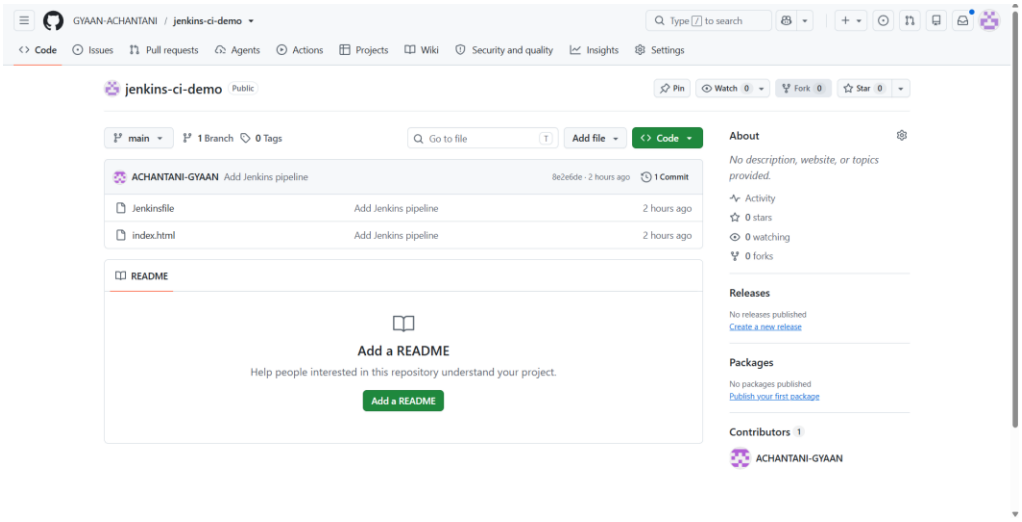
Connecting GitHub to Jenkins

GitHub is a cloud-based platform used for storing and managing source code using Git. Jenkins can be connected to GitHub so that Jenkins automatically retrieves the latest source code from a GitHub repository and performs the required build and testing operations. To connect GitHub with Jenkins, first a project or repository is created on GitHub and the project files are pushed into it. In Jenkins, a new Freestyle Project or Pipeline is then created. In the project's configuration, Git is selected as the source-code management system, and the URL of the GitHub repository is provided. If the repository is private, appropriate GitHub credentials, such as a Personal Access Token or SSH key, must be configured in Jenkins so that Jenkins has permission to access the repository.

Implementation:-




```
C:\Users\Gyaan\OneDrive\Documents\jenkins-ci-demo>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/GYAAN-ACHANTANI/jenkins-ci-demo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



Conclusion:- Thus, the Continuous Integration pipeline using Jenkins and GitHub was successfully set up and executed. The experiment demonstrated Jenkins installation and configuration, integration with a GitHub repository, and creation of a Declarative Jenkins Pipeline using a `Jenkinsfile`. The pipeline was divided into Build, Test, and Deploy stages and executed successfully, demonstrating how Jenkins can automate software development processes and support CI/CD workflows.